An Introduction to Computer Networks

Release 2.0.11

Peter L Dordal

Jul 20, 2023

CONTENTS

Preface **3**
Second Edition . . . 3
Licensing . . . 4
German Edition . . . 5
Classroom Use . . . 5
Acknowledgments . . . 7
Progress Notes . . . 8
Technical considerations . . . 9
A Note On the Cover . . . 10
Recent Changes . . . 11

1 An Overview of Networks **13**
1.1 Layers . . . 13
1.2 Data Rate, Throughput and Bandwidth . . . 14
1.3 Packets . . . 14
1.4 Datagram Forwarding . . . 16
1.5 Topology . . . 18
1.6 Routing Loops . . . 20
1.7 Congestion . . . 20
1.8 Packets Again . . . 21
1.9 LANs and Ethernet . . . 22
1.10 IP - Internet Protocol . . . 24
1.11 DNS . . . 30
1.12 Transport . . . 31
1.13 Firewalls . . . 35
1.14 Some Useful Utilities . . . 35
1.15 IETF and OSI . . . 38
1.16 Berkeley Unix . . . 41
1.17 Epilog . . . 41
1.18 Exercises . . . 41

2 Ethernet Basics **47**
2.1 10-Mbps Classic Ethernet . . . 48
2.2 100 Mbps (Fast) Ethernet . . . 59
2.3 Gigabit Ethernet . . . 60

2.4 Ethernet Switches . . . 61
2.5 Epilog . . . 65
2.6 Exercises . . . 65

3 Advanced Ethernet **69**
3.1 Spanning Tree Algorithm and Redundancy . . . 69
3.2 Virtual LAN (VLAN) . . . 74
3.3 TRILL and SPB . . . 78
3.4 Software-Defined Networking . . . 80
3.5 Epilog . . . 87
3.6 Exercises . . . 87

4 Wireless LANs **93**
4.1 Adventures in Radioland . . . 93
4.2 Wi-Fi . . . 98
4.3 WiMAX and LTE . . . 125
4.4 Fixed Wireless . . . 128
4.5 Epilog . . . 131
4.6 Exercises . . . 131

5 Other LANs **135**
5.1 Virtual Private Networks . . . 135
5.2 Carrier Ethernet . . . 136
5.3 Token Ring . . . 137
5.4 Virtual Circuits . . . 138
5.5 Asynchronous Transfer Mode: ATM . . . 142
5.6 Epilog . . . 144
5.7 Exercises . . . 144

6 Links **149**
6.1 Encoding and Framing . . . 149
6.2 Time-Division Multiplexing . . . 154
6.3 Epilog . . . 159
6.4 Exercises . . . 159

7 Packets **161**
7.1 Packet Delay . . . 161
7.2 Packet Delay Variability . . . 164
7.3 Packet Size . . . 165
7.4 Error Detection . . . 167
7.5 Epilog . . . 172
7.6 Exercises . . . 173

8 Abstract Sliding Windows **179**
8.1 Building Reliable Transport: Stop-and-Wait . . . 179
8.2 Sliding Windows . . . 184
8.3 Linear Bottlenecks . . . 187
8.4 Epilog . . . 195
8.5 Exercises . . . 195

9 IP version 4 **201**
9.1 The IPv4 Header . . . 202
9.2 Interfaces . . . 204
9.3 Special Addresses . . . 206
9.4 Fragmentation . . . 208
9.5 The Classless IP Delivery Algorithm . . . 210
9.6 IPv4 Subnets . . . 213
9.7 Network Address Translation . . . 219
9.8 Unnumbered Interfaces . . . 223
9.9 Mobile IP . . . 224
9.10 Epilog . . . 226
9.11 Exercises . . . 226

10 IPv4 Companion Protocols **229**
10.1 DNS . . . 229
10.2 Address Resolution Protocol: ARP . . . 242
10.3 Dynamic Host Configuration Protocol (DHCP) . . . 246
10.4 Internet Control Message Protocol . . . 247
10.5 Epilog . . . 251
10.6 Exercises . . . 251

11 IPv6 **253**
11.1 The IPv6 Header . . . 254
11.2 IPv6 Addresses . . . 255
11.3 Network Prefixes . . . 258
11.4 IPv6 Multicast . . . 259
11.5 IPv6 Extension Headers . . . 259
11.6 Neighbor Discovery . . . 262
11.7 IPv6 Host Address Assignment . . . 266
11.8 Epilog . . . 271
11.9 Exercises . . . 271

12 IPv6 Additional Features **273**
12.1 Globally Exposed Addresses . . . 273
12.2 ICMPv6 . . . 273
12.3 IPv6 Subnets . . . 275
12.4 Using IPv6 and IPv4 Together . . . 276
12.5 IPv6 Examples Without a Router . . . 280
12.6 IPv6 Connectivity via Tunneling . . . 283
12.7 IPv6-to-IPv4 Connectivity . . . 286
12.8 Epilog . . . 289
12.9 Exercises . . . 289

13 Routing-Update Algorithms **291**
13.1 Distance-Vector Routing-Update Algorithm . . . 292
13.2 Distance-Vector Slow-Convergence Problem . . . 296
13.3 Observations on Minimizing Route Cost . . . 298
13.4 Loop-Free Distance Vector Algorithms . . . 300

13.5 Link-State Routing-Update Algorithm . . . 306
13.6 Routing on Other Attributes . . . 310
13.7 ECMP . . . 312
13.8 Epilog . . . 313
13.9 Exercises . . . 313

14 Large-Scale IP Routing **319**
14.1 Classless Internet Domain Routing: CIDR . . . 319
14.2 Hierarchical Routing . . . 321
14.3 Legacy Routing . . . 322
14.4 Provider-Based Routing . . . 323
14.5 Geographical Routing . . . 328
14.6 Epilog . . . 329
14.7 Exercises . . . 329

15 Border Gateway Protocol (BGP) **333**
15.1 AS-paths . . . 334
15.2 AS-Paths and Route Aggregation . . . 336
15.3 Transit Traffic . . . 337
15.4 BGP Filtering and Routing Policies . . . 337
15.5 BGP Table Size . . . 339
15.6 BGP Path attributes . . . 340
15.7 BGP and Traffic Engineering . . . 344
15.8 BGP and Anycast . . . 347
15.9 BGP for Interior Routing . . . 347
15.10 BGP Relationships . . . 348
15.11 Examples of BGP Instability . . . 352
15.12 BGP Security and Route Registries . . . 354
15.13 Epilog . . . 357
15.14 Exercises . . . 358

16 UDP Transport **361**
16.1 User Datagram Protocol – UDP . . . 361
16.2 Trivial File Transport Protocol, TFTP . . . 373
16.3 Fundamental Transport Issues . . . 375
16.4 Other TFTP notes . . . 380
16.5 Remote Procedure Call (RPC) . . . 383
16.6 Epilog . . . 388
16.7 Exercises . . . 388

17 TCP Transport Basics **393**
17.1 The End-to-End Principle . . . 394
17.2 TCP Header . . . 394
17.3 TCP Connection Establishment . . . 396
17.4 TCP and WireShark . . . 400
17.5 TCP Offloading . . . 402
17.6 TCP simplex-talk . . . 402
17.7 TCP and `bind()` . . . 407

17.8 TCP state diagram . . . 408
17.9 Epilog . . . 414
17.10 Exercises . . . 414

18 TCP Issues and Alternatives **419**
18.1 TCP Old Duplicates . . . 419
18.2 TIMEWAIT . . . 420
18.3 The Three-Way Handshake Revisited . . . 421
18.4 Anomalous TCP scenarios . . . 423
18.5 TCP Faster Opening . . . 424
18.6 Path MTU Discovery . . . 425
18.7 TCP Sliding Windows . . . 426
18.8 TCP Delayed ACKs . . . 426
18.9 Nagle Algorithm . . . 427
18.10 TCP Flow Control . . . 428
18.11 Silly Window Syndrome . . . 428
18.12 TCP Timeout and Retransmission . . . 429
18.13 KeepAlive . . . 430
18.14 TCP timers . . . 431
18.15 Variants and Alternatives . . . 431
18.16 Epilog . . . 441
18.17 Exercises . . . 441

19 TCP Reno and Congestion Management **443**
19.1 Basics of TCP Congestion Management . . . 444
19.2 Slow Start . . . 448
19.3 TCP Tahoe and Fast Retransmit . . . 453
19.4 TCP Reno and Fast Recovery . . . 454
19.5 TCP NewReno . . . 458
19.6 Selective Acknowledgments (SACK) . . . 460
19.7 TCP and Bottleneck Link Utilization . . . 461
19.8 Single Packet Losses . . . 464
19.9 TCP Assumptions and Scalability . . . 465
19.10 TCP Parameters . . . 466
19.11 Epilog . . . 466
19.12 Exercises . . . 467

20 Dynamics of TCP **471**
20.1 A First Look At Queuing . . . 471
20.2 Bottleneck Links with Competition . . . 472
20.3 TCP Reno Fairness with Synchronized Losses . . . 480
20.4 Epilog . . . 487
20.5 Exercises . . . 487

21 Further Dynamics of TCP **493**
21.1 Notions of Fairness . . . 493
21.2 TCP Reno loss rate versus `cwnd` . . . 495
21.3 TCP Friendliness . . . 497

21.4 AIMD Revisited . . . 499
21.5 Active Queue Management . . . 501
21.6 The High-Bandwidth TCP Problem . . . 506
21.7 The Lossy-Link TCP Problem . . . 508
21.8 The Satellite-Link TCP Problem . . . 508
21.9 Epilog . . . 509
21.10 Exercises . . . 509

22 Newer TCP Implementations **515**
22.1 Choosing a TCP on Linux . . . 515
22.2 High-Bandwidth Desiderata . . . 518
22.3 RTTs . . . 519
22.4 A Roadmap . . . 519
22.5 Highspeed TCP . . . 519
22.6 TCP Vegas . . . 522
22.7 FAST TCP . . . 525
22.8 TCP Westwood . . . 527
22.9 TCP Illinois . . . 529
22.10 Compound TCP . . . 530
22.11 TCP Veno . . . 532
22.12 TCP Hybla . . . 533
22.13 DCTCP . . . 533
22.14 H-TCP . . . 536
22.15 TCP CUBIC . . . 537
22.16 TCP BBR . . . 542
22.17 Epilog . . . 546
22.18 Exercises . . . 547

23 Queuing and Scheduling **553**
23.1 Queuing and Real-Time Traffic . . . 554
23.2 Traffic Management . . . 554
23.3 Priority Queuing . . . 555
23.4 Queuing Disciplines . . . 555
23.5 Fair Queuing . . . 556
23.6 Applications of Fair Queuing . . . 569
23.7 Hierarchical Queuing . . . 571
23.8 Hierarchical Weighted Fair Queuing . . . 574
23.9 Epilog . . . 580
23.10 Exercises . . . 580

24 Token Bucket Rate Limiting **585**
24.1 Token Bucket Definition . . . 586
24.2 Token-Bucket Examples . . . 588
24.3 Multiple Token Buckets . . . 589
24.4 GCRA . . . 590
24.5 Guaranteeing VoIP Bandwidth . . . 591
24.6 Limiting Delay . . . 592
24.7 Token Bucket Through One Router . . . 593

24.8 Token Bucket Through Multiple Routers . . . 594
24.9 Delay Constraints . . . 594
24.10 CBQ . . . 597
24.11 Linux HTB . . . 597
24.12 Parekh-Gallager Theorem . . . 598

25 Quality of Service **603**
25.1 Net Neutrality . . . 604
25.2 Where the Wild Queues Are . . . 604
25.3 Real-time Traffic . . . 605
25.4 Integrated Services / RSVP . . . 607
25.5 Global IP Multicast . . . 608
25.6 RSVP . . . 613
25.7 Differentiated Services . . . 617
25.8 RED with In and Out . . . 621
25.9 NSIS . . . 622
25.10 Comcast Congestion-Management System . . . 622
25.11 Real-time Transport Protocol (RTP) . . . 623
25.12 Multi-Protocol Label Switching (MPLS) . . . 628
25.13 Epilog . . . 630
25.14 Exercises . . . 630

26 Network Management and SNMP **633**
26.1 Network Architecture . . . 635
26.2 SNMP Basics . . . 635
26.3 SNMP Naming and OIDs . . . 637
26.4 MIBs . . . 639
26.5 SNMPv1 Data Types . . . 640
26.6 ASN.1 Syntax and SNMP . . . 640
26.7 SNMP Tables . . . 642
26.8 SNMP Operations . . . 646
26.9 MIB Browsing . . . 651
26.10 MIB-2 . . . 652
26.11 SNMPv1 communities and security . . . 661
26.12 SNMP and ASN.1 Encoding . . . 662
26.13 Network Management Systems . . . 665
26.14 SNMP Alternatives . . . 665
26.15 Exercises . . . 666

27 SNMP versions 2 and 3 **669**
27.1 SNMPv2 . . . 669
27.2 Table Row Creation . . . 680
27.3 SNMPv3 . . . 689
27.4 Exercises . . . 699

28 Security **701**
28.1 Code-Execution Intrusion . . . 702
28.2 Stack Buffer Overflow . . . 703

28.3 Heap Buffer Overflow . . . 712
28.4 Network Intrusion Detection . . . 717
28.5 Cryptographic Goals . . . 718
28.6 Secure Hashes . . . 719
28.7 Shared-Key Encryption . . . 724
28.8 Diffie-Hellman-Merkle Exchange . . . 733
28.9 Exercises . . . 737

29 Public-Key Encryption **739**
29.1 RSA . . . 739
29.2 Forward Secrecy . . . 742
29.3 Trust and the Man in the Middle . . . 743
29.4 End-to-End Encryption . . . 744
29.5 SSH and TLS . . . 744
29.6 IPsec . . . 763
29.7 DNSSEC . . . 766
29.8 RSA Key Examples . . . 775
29.9 Exercises . . . 778

30 Mininet **781**
30.1 Installing Mininet . . . 782
30.2 A Simple Mininet Example . . . 784
30.3 Multiple Switches in a Line . . . 786
30.4 IP Routers in a Line . . . 789
30.5 IP Routers With Simple Distance-Vector Implementation . . . 791
30.6 Quagga Routing and BGP . . . 794
30.7 TCP Competition . . . 799
30.8 Linux Traffic Control (tc) . . . 809
30.9 OpenFlow and the POX Controller . . . 812
30.10 Exercises . . . 824

31 Network Simulations: ns-2 **827**
31.1 The ns-2 simulator . . . 827
31.2 A Single TCP Sender . . . 829
31.3 Two TCP Senders Competing . . . 841
31.4 TCP Loss Events and Synchronized Losses . . . 857
31.5 TCP Reno versus TCP Vegas . . . 866
31.6 Wireless Simulation . . . 868
31.7 Epilog . . . 874
31.8 Exercises . . . 874

32 The ns-3 Network Simulator **877**
32.1 Installing and Running ns-3 . . . 877
32.2 A Single TCP Sender . . . 878
32.3 Wireless . . . 887
32.4 Exercises . . . 893

33 Bibliography **895**